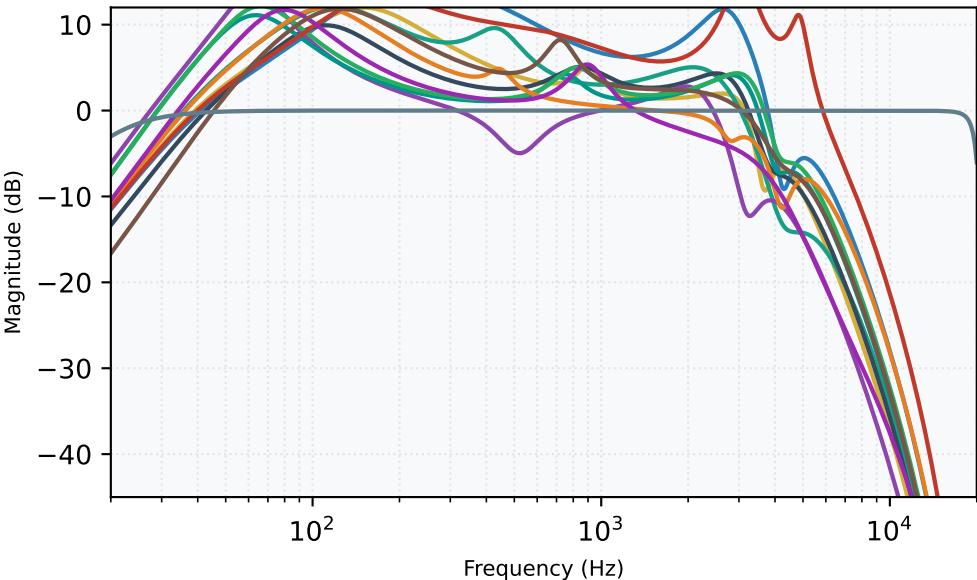


IRONSTACK — ACOUSTIC CABINET ENCLOSURE MODELS BENCHMARK

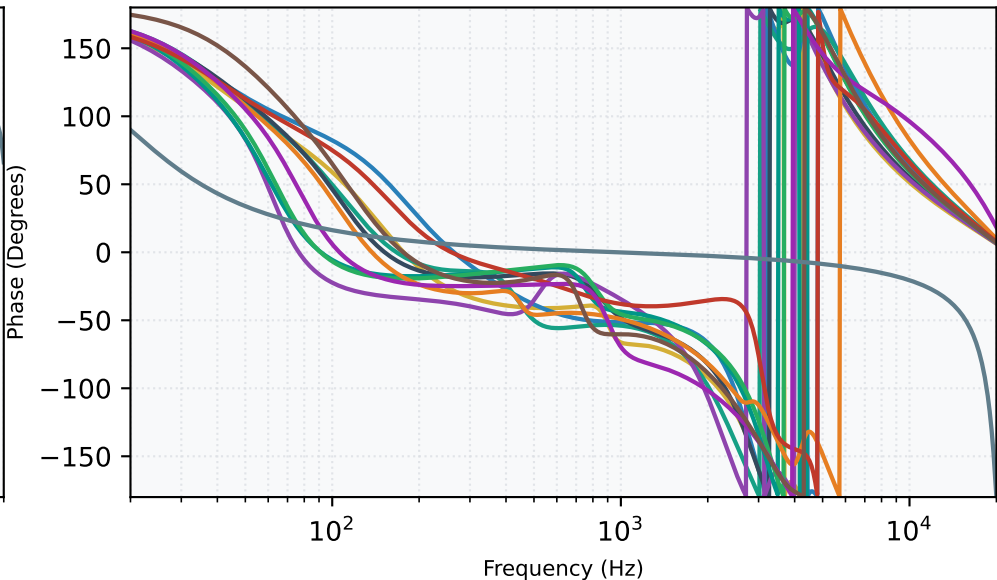
Frequency & Phase Transfer Functions across 12 Discrete Biquad Cascade Enclosures

Zero-ML / Zero-Latency

Cabinet Enclosure Frequency Response



Cabinet Phase Responses



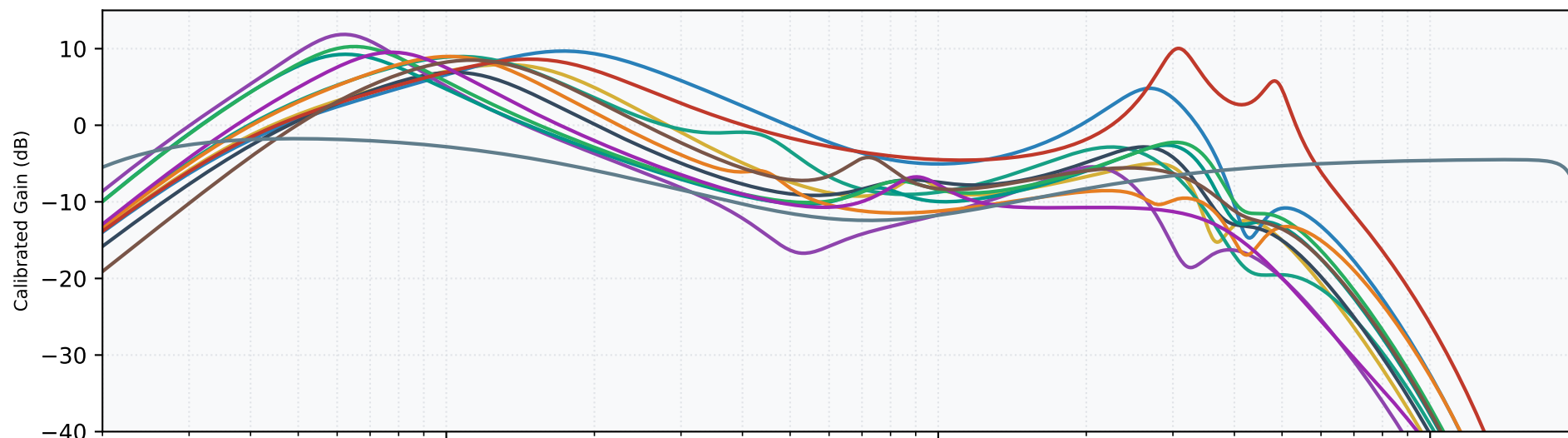
Cabinet Model	Enclosure Type	Thump Freq	Thump Boost	Mid Voicing	Presence Peak	Roll-off	Acoustic Sonic Character
4x10 Bassman Neo (Tone3000)	Open Pine	149.5 Hz	+12.0 dB	900 Hz (+3.5dB)	3373 Hz (+7.7dB)	4000 Hz	Punchy fast transient bass articulation
2x12 Twin C12N (Tone3000)	Open Birch	180.0 Hz	+12.0 dB	443 Hz (+6.0dB)	2750 Hz (+12.4dB)	5139 Hz	Deep Blackface lows, singing glassy bell
2x15 '70 Bassman CTS (Tone3000)	Deep Sealed	116.0 Hz	+12.0 dB	440 Hz (+6.0dB)	2423 Hz (+7.2dB)	4408 Hz	Sub-bass authority, massive physical kick
8x10 Ampeg SVT Fridge	Infinite Baffle	110.0 Hz	+10.0 dB	850 Hz (+4.2dB)	2850 Hz (+6.8dB)	4200 Hz	Stadium low-mid punch, tight transient slam
1x18 Acoustic 360 Horn	Folded Horn	62.0 Hz	+14.0 dB	520 Hz (-5.5dB)	2150 Hz (+5.0dB)	3600 Hz	Jaco sub-bass acoustic compression throw
2x10 Eminence Legend (Vented)	Vented 3 cu.ft	62.0 Hz	+11.5 dB	750 Hz (+3.8dB)	3200 Hz (+7.5dB)	4500 Hz	Lucas vented design, 62Hz Helmholtz tuning
4x10 Eminence Legend (Vented)	Vented 6 cu.ft	65.0 Hz	+12.5 dB	820 Hz (+4.5dB)	3400 Hz (+8.2dB)	4600 Hz	Lucas vented design, 65Hz massive punch
2x12 Hartke Pro 2200 (Tone3000)	Ported Dual	108.0 Hz	+12.0 dB	447 Hz (+2.5dB)	2796 Hz (-2.3dB)	5210 Hz	Aluminum cone lightning slap & transient
4x12 Marshall 1960A (Tone3000)	Closed 1960	161.0 Hz	+12.0 dB	689 Hz (+6.0dB)	3073 Hz (+15.0dB)	6153 Hz	Creamy British roar, forward aggressive bite
4x12 Mesa Recto V30 (Tone3000)	Closed Oversized	78.4 Hz	+12.0 dB	900 Hz (+6.0dB)	3800 Hz (+3.9dB)	4915 Hz	Tight modern low thump & scooped bite
2x12 '66 Bassman C12NA (Tone3000)	Closed Vintage	121.3 Hz	+12.0 dB	724 Hz (+6.0dB)	2632 Hz (+2.7dB)	4521 Hz	Warm vintage acoustic resonance, rich mids
Bypass (Direct Out)	Direct Line	Flat	0.0 dB	Flat	Flat	20.0 kHz	Flat direct signal for external IR loader

IRONSTACK — END-TO-END RIG RESPONSE MATRIX (PART 1 OF 5)

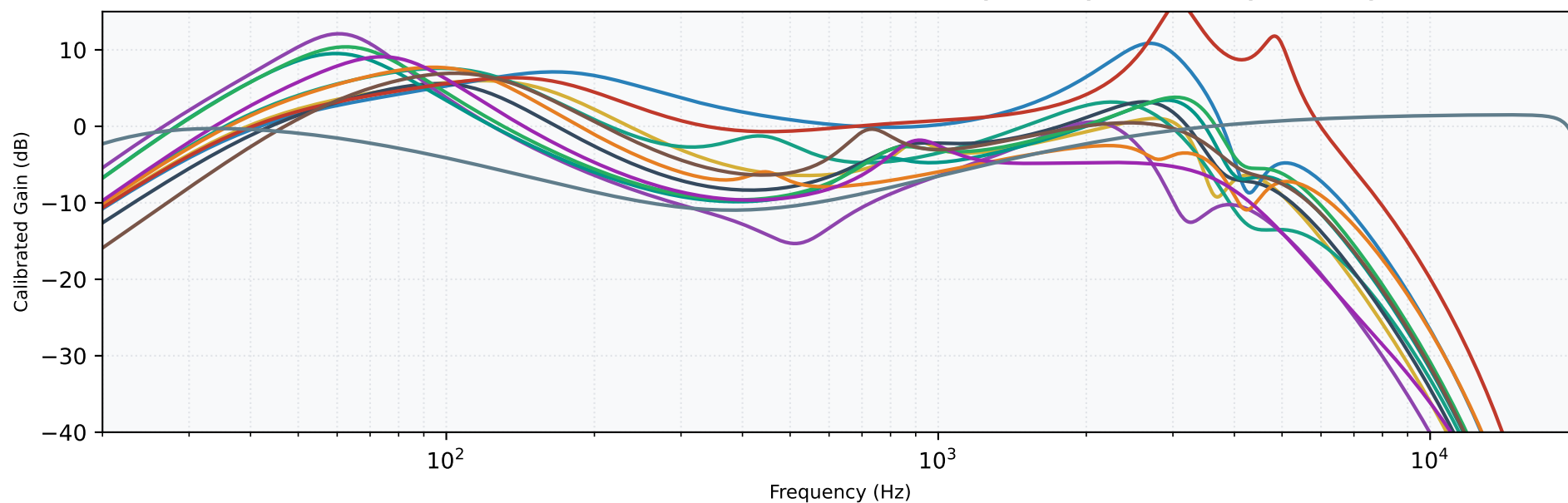
Calibrated Output Response across 12 Cabinet Models with Inter-Amp Loudness Normalization

Rigs 1-24 of 120

'59 Fender Bassman (5F6-A) (T=5, M=5, B=5) — Original Tweed benchmark; balanced warmth, 800Hz scoop [Norm: +0.0 dB]



'65 Fender Bassman (AA864) (T=5, M=5, B=5) — Blackface Bassman; 100nF deep bass cap, 47nF mid cap, 100k slope [Norm: +6.5 dB]



Cabinet Models (12 Discrete Enclosures)

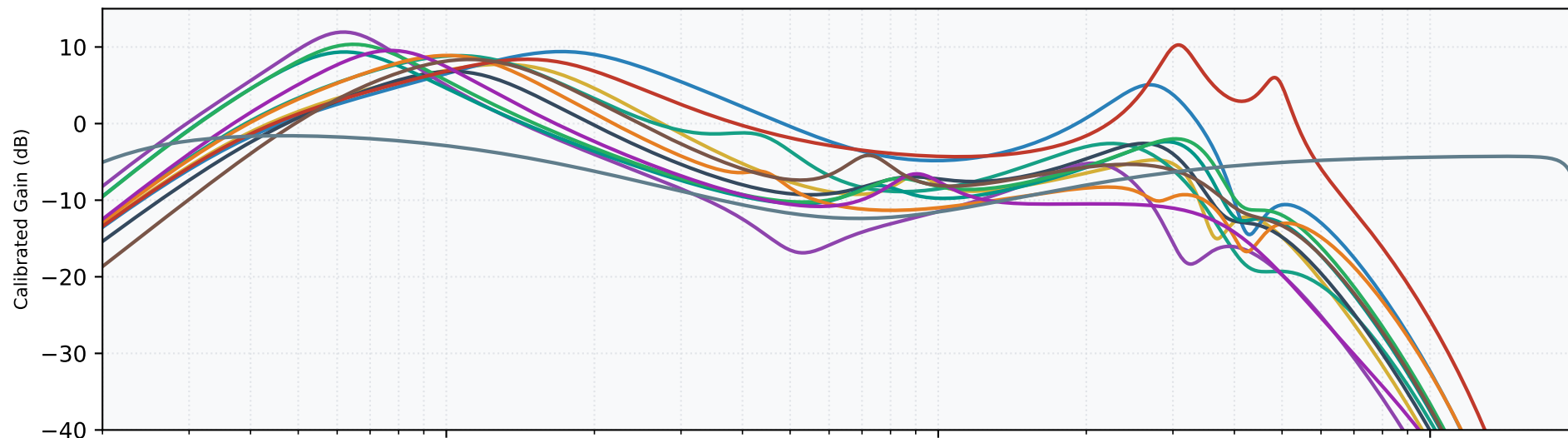


IRONSTACK — END-TO-END RIG RESPONSE MATRIX (PART 2 OF 5)

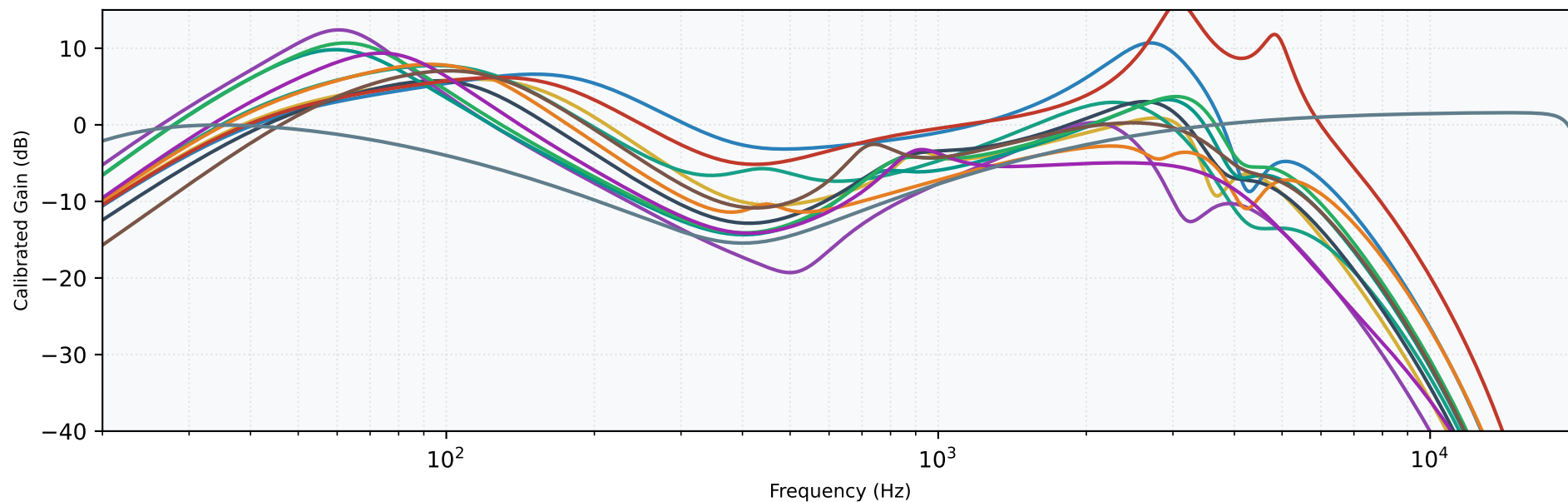
Calibrated Output Response across 12 Cabinet Models with Inter-Amp Loudness Normalization

Rigs 25-48 of 120

Marshall Super Bass 100 (JMP 1992) (T=5, M=5, B=5) — British heavy tube bass; 56k slope, 250pF treble, 22nF bass/mid, aggressive push [Norm: +0.2 dB]



Fender Twin Reverb (AB763) (T=5, M=5, B=5) — Classic Blackface; huge low-end authority, deeper scoop at 450Hz [Norm: +7.2 dB]



Cabinet Models (12 Discrete Enclosures)

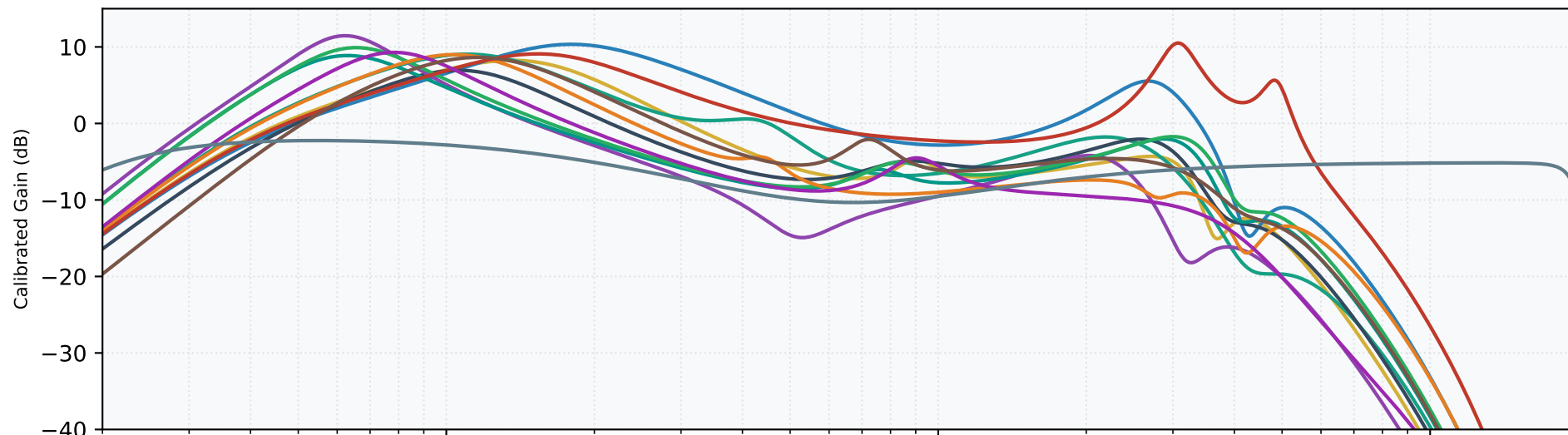


IRONSTACK — END-TO-END RIG RESPONSE MATRIX (PART 3 OF 5)

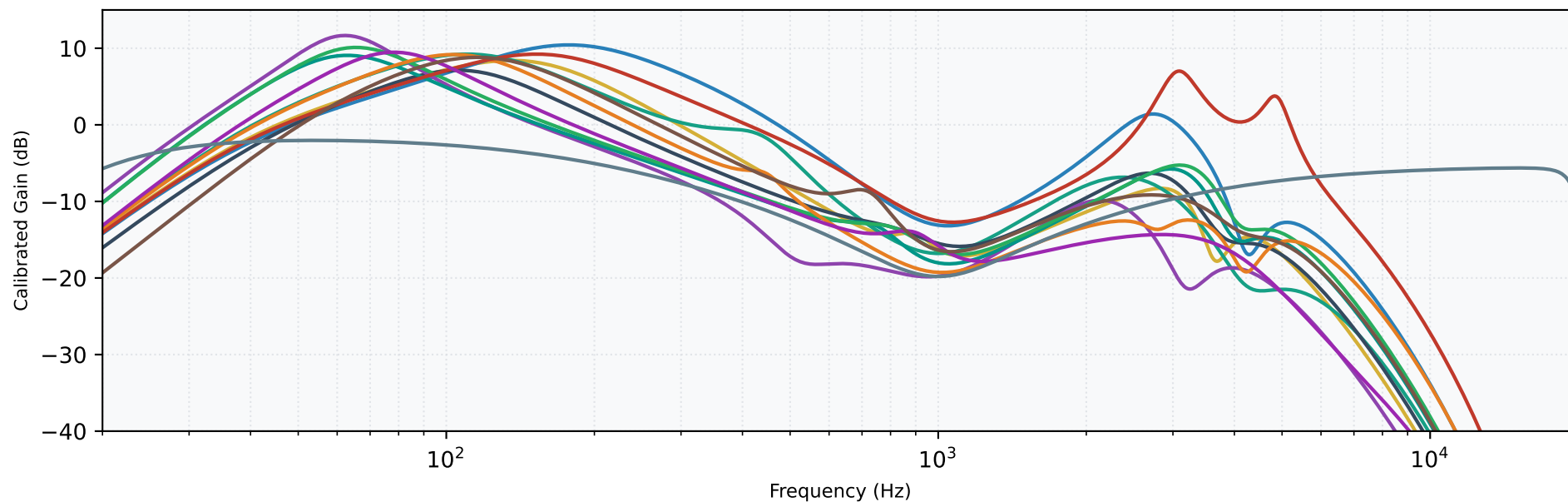
Calibrated Output Response across 12 Cabinet Models with Inter-Amp Loudness Normalization

Rigs 49-72 of 120

Marshall JCM800 / 1959 Plexi (T=5, M=5, B=5) — British crunch; forward midrange, higher treble bite, 33k slope [Norm: -1.2 dB]



Vox AC30 Top Boost (T=5, M=5, B=5) — British chime; 47pF small treble cap, 100k slope, sparkling highs [Norm: +0.2 dB]



Cabinet Models (12 Discrete Enclosures)

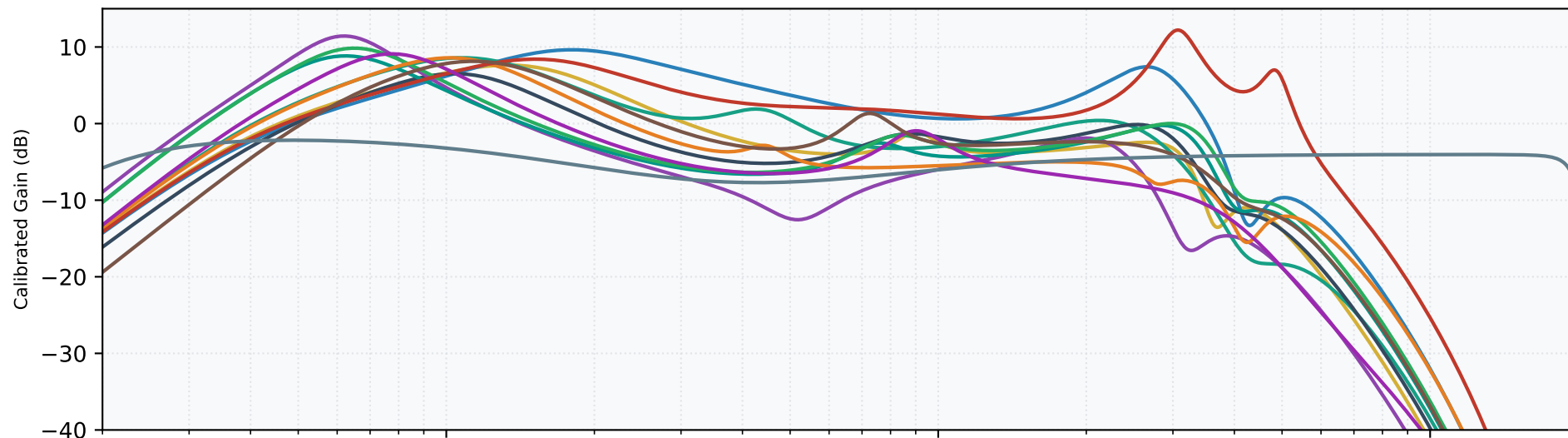


IRONSTACK — END-TO-END RIG RESPONSE MATRIX (PART 4 OF 5)

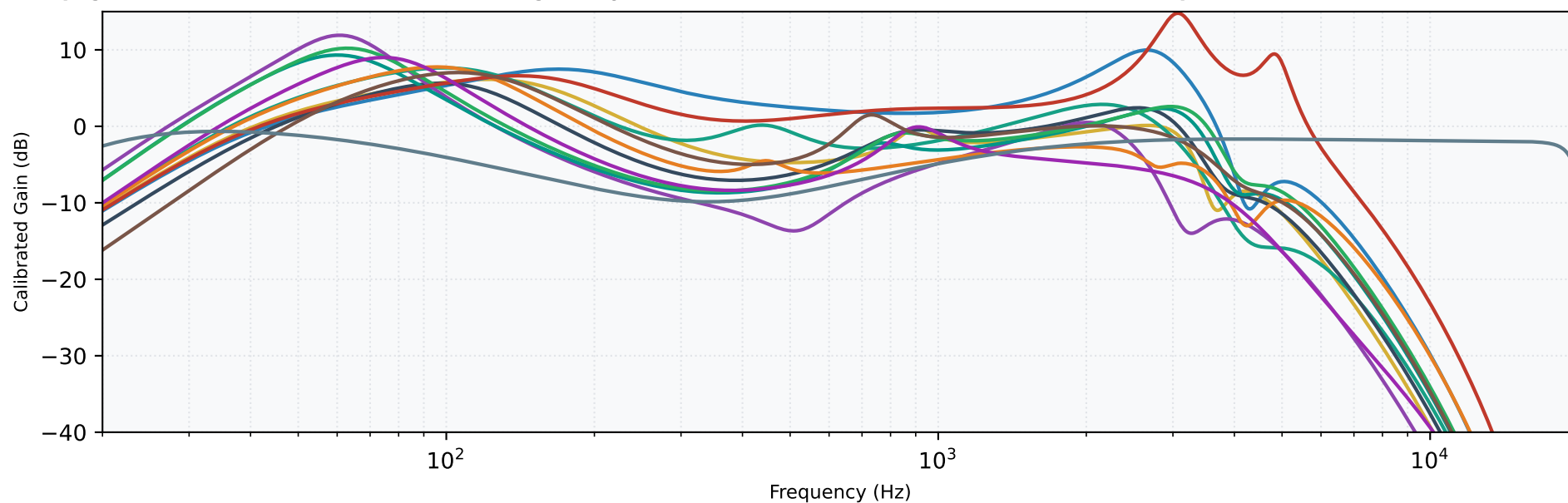
Calibrated Output Response across 12 Cabinet Models with Inter-Amp Loudness Normalization

Rigs 73-96 of 120

Mesa Boogie Dual Rectifier (T=5, M=5, B=5) — Modern metal; 680pF treble cap, 50k mid pot for extreme scoop-to-punch [Norm: -0.8 dB]



Ampeg B-15N Portaflex (T=5, M=5, B=5) — Legendary tube bass James/Baxandall stack; flat center, independent +/-12dB shelves [Norm: +11.2 dB]



Cabinet Models (12 Discrete Enclosures)

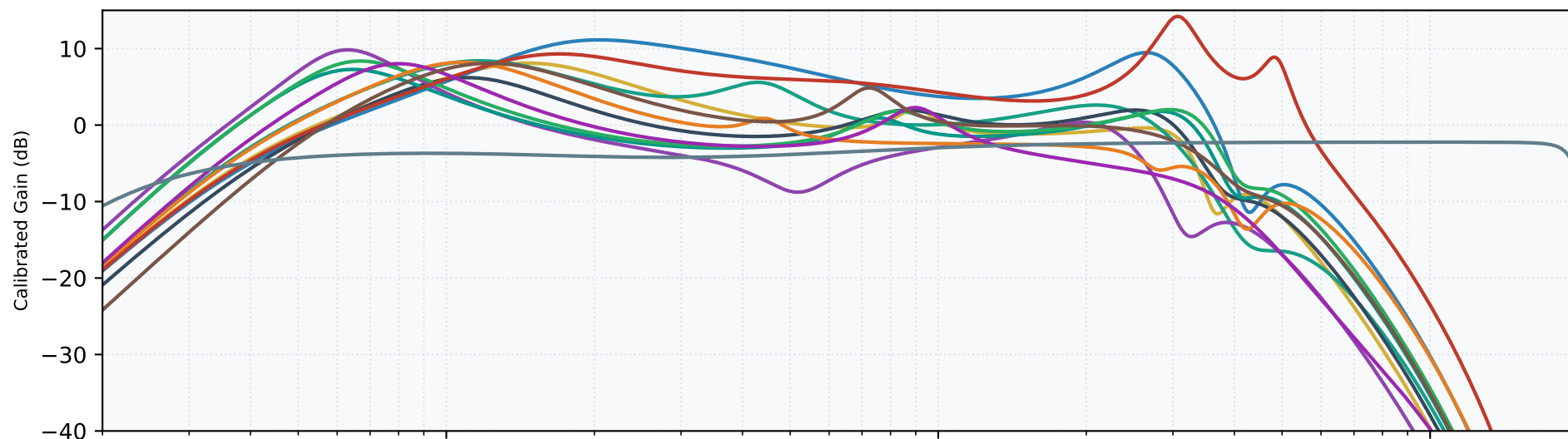


IRONSTACK — END-TO-END RIG RESPONSE MATRIX (PART 5 OF 5)

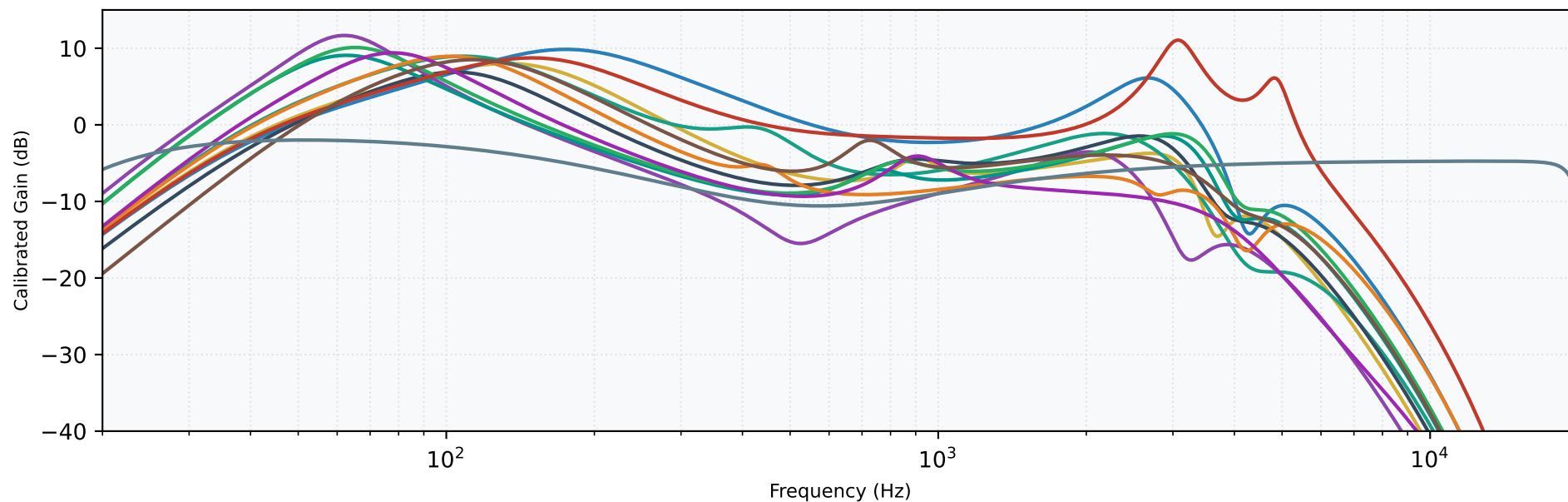
Calibrated Output Response across 12 Cabinet Models with Inter-Amp Loudness Normalization

Rigs 97-120 of 120

Ampeg B-100R Rocket Bass (T=5, M=5, B=5) — Iconic solid-state/tube emulation combo with Ultra-Mid contour; 50k pots, 33k slope, 4.7nF treble [Norm: -0.0]



Soldano SLO-100 (T=5, M=5, B=5) — Boutique high-gain; legendary singing lead, smooth transition [Norm: -0.5 dB]



Cabinet Models (12 Discrete Enclosures)

